

$$2^2 \times 3 \times 5 = 4 \times 3 \times 5 = 60$$

Ans

All lights will flash together after ~~after~~ 60 seconds.

Day - 12

Date: 19-01-26

Recurring decimal to fraction

Recurring decimal

- Type 1
- Type 2
- Type 3

Type 1 (short cuts)

$$1) 0.\bar{7} = \frac{7}{9}$$

$$2) 0.\bar{35} = \frac{35}{99}$$

Notes (Base)

$$x = 0.\bar{7} \Rightarrow 0.7777 \dots$$

$$10x = 7.7777 \dots$$

$$-x = 0.7777$$

$$9x = 7$$

$$x = \frac{7}{9}$$

$$x = 0.3535 \dots$$

$$100x = 35.3535 \dots$$

$$-x = 0.3535$$

$$99x = 35$$

$$x = \frac{35}{99}$$

$$3) 0.\overline{71} = \frac{71}{99}$$

$$4) 0.\overline{562} = \frac{562}{999}$$

$$5) 0.\overline{712} = \frac{712}{999}$$

$$x = 0.562562$$

$$1000x = 562.562$$

$$-x = 0.562562$$

$$999x = 562$$

$$x = \frac{562}{999}$$

TYPE - 2 (short cuts)

$$1) 1.\overline{34} = \frac{134-1}{99} = \frac{133}{99}$$

$$2) 2.\overline{5} = \frac{25-2}{9} = \frac{23}{9}$$

$$3) 3.\overline{537} = \frac{3537-3}{999} = \frac{3534}{999}$$

$$4) 2.\overline{82} = \frac{282-2}{99} = \frac{280}{99}$$

Notes (Base)

$$x = 1.\overline{34} \Rightarrow 1.3434 \dots$$

$$100x = 134.3434$$

$$-x = 1.3434 \dots$$

$$99x = 133$$

$$x = \frac{133}{99}$$

$$x = \overline{3.537} = 3.537537 \dots$$

$$1000x = 3537.537537 \dots$$

$$-x = \underline{\underline{00003.537537}}$$

$$999x = 3534 \text{ ~~537537~~}$$

$$x = \frac{3534}{999}$$

TYPE 3 (short cuts)

$$1) 1.2\overline{3} = \frac{123 - 12}{90} = \frac{111}{90}$$

$$2) 4.3\overline{67} \Rightarrow \frac{4367 - 43}{990} = \frac{4324}{990}$$

$$3) 2.5\overline{89} \Rightarrow \frac{2589 - 25}{990} \Rightarrow \frac{2564}{990}$$

Notes (Base)

$$x = 4.3\overline{67} \Rightarrow 4.367367 \dots$$

$$1000x = 4367.676767 \dots$$

$$-10x = \underline{\underline{43.676767 \dots}}$$

$$990x = 4324$$

$$x = \frac{4324}{990}$$

practice Exercises

$$1) 0.\overline{3} =)$$

$$10x = 3.33\ldots$$

$$-x = 0.33\ldots$$

$$9x = 3$$

$$x = \frac{3}{9}$$

$$2) 0.\overline{142857} =) \quad \begin{array}{r} 142857 \\ \hline 999999 \end{array}$$